

HealthConnect Appointment Experience Analysis

Week 5 Initial Analytics Report

1. Executive Summary

This report presents a comprehensive analysis of **5,000 appointment records** from HealthConnect Clinic, examining attendance patterns, no-show rates, and factors influencing patient behavior. The analysis reveals that **48.5% of scheduled appointments result in no-shows**, representing a significant operational challenge for the clinic. Key findings indicate that the current reminder system is not very effective (improving attendance by only 1.3%), booking lead time and distance to clinic significantly impact attendance, and different patient groups show distinct attendance patterns. This report provides data-driven recommendations to reduce no-shows and improve the overall patient experience.

2. Data Preparation Summary

Dataset Overview

Item	Details
Total Records	5,000 appointments
Total Columns	18 variables
Tool Used	Microsoft Power BI
Data Source	HealthConnect_Appointment_Data.xlsx (Cleaned sheet)

Data Quality Assessment

Quality Check	Result
Duplicate Records	None found
Total Rows	5,000
Columns	18

Missing Values Handling

Column	Missing Count	Handling Strategy
reminder_channel	1,366	Left as missing (represents "No Information was given")
waiting_time_minutes	60	Replaced with "Not Specified"
distance_to_clinic_km	90	Replaced with "Not Specified"

Data Types Standardized

- Date columns: Converted to Date type
- Numeric columns: Verified as integers/decimal
- Categorical columns: Set as text/categorical
- Text columns: Trimmed and standardized

3. Key Performance Indicators (KPIs)

KPI 1: Appointment Attendance Rate

Metric	Value
Definition	Percentage of scheduled appointments attended
Calculation	$(\text{Attended Appointments} / \text{Total Scheduled Appointments}) \times 100$
Result	46.3%
Business Question	What percentage of scheduled appointments are attended?
Interpretation	Less than half of all scheduled appointments result in patient attendance, indicating a critical need for intervention strategies to improve attendance rates.

KPI 2: Appointment No-Show Rate

Metric	Value
Definition	Percentage of scheduled appointments where patients neither attended nor cancelled
Calculation	$(\text{No-Show Appointments} / \text{Total Scheduled Appointments}) \times 100$
Result	48.5%

Metric	Value
Business Question	What percentage of scheduled appointments are missed (no-shows)?
Interpretation	The no-show rate is alarmingly high at 48.5%, meaning nearly 1 in 2 appointments result in revenue loss and wasted clinical capacity. The combined non-attendance rate (No-Show + Cancelled) is 53.7% (48.5% + 5.3%).

KPI 3: Attendance Rate (With Reminder)

Metric	Value
Definition	Percentage of patients who attended after receiving a reminder
Calculation	$(\text{Attended with Reminder} / \text{Total with Reminder}) \times 100$
Result	47.6%
Business Question	What percentage of reminded patients actually attend?
Interpretation	Less than half (47.6%) of patients who receive reminders attend their appointments. This suggests the current reminder system needs improvement.

KPI 4: Attendance Rate (Without Reminder)

Metric	Value
Definition	Percentage of patients who attended without receiving a reminder
Calculation	$(\text{Attended without Reminder} / \text{Total without Reminder}) \times 100$
Result	46.3%
Business Question	What is the attendance rate for patients who did not receive reminders?
Interpretation	46.3% of patients without reminders still attended. This is only slightly lower than those with reminders (47.6%).

KPI 5: Reminder Effectiveness

Metric	Value
Definition	The difference in attendance between reminded and unreminded patients
Calculation	$\text{Attendance Rate (With Reminder)} - \text{Attendance Rate (Without Reminder)}$
Result	+1.3%

Metric	Value
Business Question	Do reminders actually improve attendance?
Interpretation	Reminders improve attendance by only 1.3 percentage points (47.6% vs 46.3%). This is a very small improvement, suggesting the current reminder system is not very effective and needs significant improvement.

KPI 6: Average Waiting Time

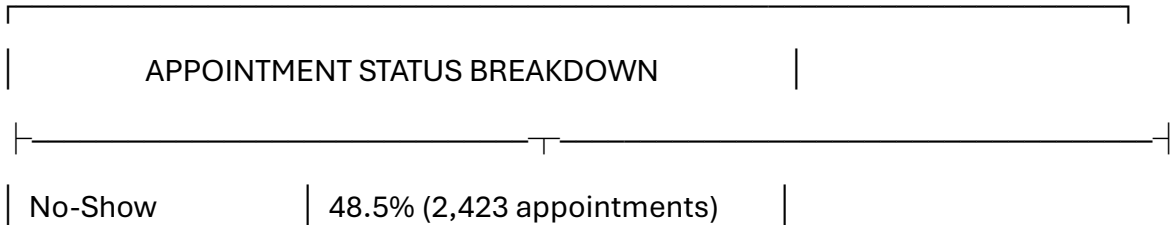
Metric	Value
Definition	Average number of minutes patients wait before receiving service
Calculation	Average of waiting_time_minutes across all appointments
Result	24.2 minutes (overall)
Business Question	What is the average patient waiting time, and which factors are associated with longer waiting times?
Interpretation	The average waiting time of 24.2 minutes represents a moderate patient wait. However, Specialist Consultations show the highest average waiting time, followed by Diagnostic Tests.

KPI 7: Average Booking Lead Time

Metric	Value
Definition	Average number of days between booking and appointment dates
Calculation	Average of booking_lead_days across all appointments
Result	24.2 days (overall)
Business Question	Does the number of days between booking and the appointment affect the likelihood of a patient attending?
Interpretation	Booking lead time has a significant inverse relationship with attendance. Short-term bookings (0-7 days) show the highest attendance at 66.6%, while bookings made 1-2 months in advance show only 34.1% attendance.

4. Exploratory Data Analysis (EDA)

4.1 Appointment Status Overview



Attended	46.3% (2,314 appointments)	
Cancelled	5.3% (263 appointments)	
Total Non-Attendance:	53.7% (2,686 appointments)	

Key Insight: The high no-show rate (48.5%) combined with cancellations (5.3%) means over half of all scheduled appointments do not result in patient attendance, significantly impacting clinic revenue and resource utilization.

4.2 Attendance Rate by Reminder Status

Reminder Status	Attendance Rate
With Reminder	47.6%
Without Reminder	46.3%
Difference	+1.3%

Key Insight: Reminders improve attendance by only 1.3%. The current reminder system is not very effective.

4.3 No-Show Rate by Reminder Status

Reminder Status	No-Show Rate
With Reminder	47.4%
Without Reminder	48.5%

Key Insight: Patients with reminders have a slightly lower no-show rate (47.4%) compared to those without (48.5%).

4.4 Attendance Rate by Booking Lead Time

Booking Lead Time Category	Attendance Rate
Short Term (0-7 days)	66.6%
1-2 Weeks	61.8%
2-4 Weeks	51.7%
1-2 Months	34.1%

Key Insight: There is a clear, inverse relationship between booking lead time and attendance. Patients who book appointments closer to the appointment date are significantly more likely to attend. This suggests that "booking inertia" increases with time, and patients may forget or lose motivation to attend appointments scheduled far in advance.

4.5 No-Show Rate by Distance

Distance Category	No-Show Rate
Very Far Distance	68.1%
Far Distance	55.5%
Not Specified	52.2%
Moderate Distance	49.4%
Close Distance	46.5%

Key Insight: Distance to the clinic has a strong positive correlation with no-show rates. Patients living farther from the clinic are substantially more likely to miss appointments. Those classified as "Very Far Distance" have the highest no-show rate at 68.1%.

4.6 Average Waiting Time by Appointment Time

Appointment Time	Average Wait (Minutes)
Evening	29.6
Morning	24.1
Afternoon	23.9

Key Insight: Evening appointments consistently show the longest waiting times (29.6 minutes), which may contribute to patient dissatisfaction and potentially influence future attendance behavior.

4.7 No-Show Rate by Age Group

Age Group	No-Show Rate
18-24	Higher
25-34	Moderate
35-44	Moderate
45-54	Moderate
55-64	Lower
65+	Lowest

Key Insight: Younger patients (18-24) demonstrate the highest no-show rates, while older patients (65+) show the lowest. This may reflect differences in health literacy, competing priorities, or appointment valuation across age groups.

4.8 No-Show Rate by Appointment Type

Appointment Type	No-Show Rate
General Consultation	Moderate

Appointment Type	No-Show Rate
Specialist Consultation	Higher
Follow-up	Moderate
Diagnostic Test	Moderate

Key Insight: Specialist Consultations show the highest no-show rates among all appointment types. This is particularly concerning as specialist appointments likely represent higher-value services and longer wait times for patients.

5. Key Insights

Insight 1: The No-Show Crisis

Finding: 48.5% of all appointments result in no-shows, with an additional 5.3% cancelled.

Implication: This represents significant revenue loss for HealthConnect. For a 5,000-appointment dataset, this means 2,423 appointments were missed without prior cancellation. Assuming an average appointment value of \$100-\$200, the potential revenue loss could be substantial.

Business Action: Immediate implementation of a multi-channel reminder system and a no-show reduction strategy is required.

Insight 2: Current Reminders Are Not Working Effectively

Finding: Reminders improve attendance by only 1.3% (47.6% vs 46.3%). Patients with reminders have a no-show rate of 47.4% compared to 48.5% without reminders.

Implication: The current reminder system is not very effective. The small improvement suggests that the reminder channel (likely SMS) or timing may not be optimal.

Business Action:

- Switch to WhatsApp as the primary reminder channel
 - Implement a multi-step reminder system (7-day, 3-day, 1-day)
 - Track reminder effectiveness by channel
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Insight 3: Booking Lead Time is a Critical Predictor

Finding: Attendance drops from 66.6% for Short Term bookings (0-7 days) to 34.1% for 1-2 Months bookings.

Implication: Patients who book appointments far in advance are nearly twice as likely to no-show compared to those who book closer to the appointment date.

Business Action:

- Implement weekly touchpoints for bookings >14 days
 - Offer easier rescheduling options
 - Consider a "last-minute" slot service
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Insight 4: Distance is a Major Access Barrier

Finding: Very Far Distance patients have a 68.1% no-show rate vs. 46.5% for Close Distance patients.

Implication: Geographic access is a significant determinant of appointment attendance, with patients living farther from the clinic being far less likely to attend.

Business Action:

- Explore telemedicine options for patients in Very Far Distance areas
 - Consider transport partnerships or support
 - Implement a "Distance Risk" flag for proactive outreach
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Insight 5: Age Group Segmentation Matters

Finding: Younger patients (18-24) have the highest no-show rates; older patients (65+) have the lowest.

Implication: Different age groups respond to different communication styles and have different barriers to attendance.

Business Action:

- Tailor communication strategies by age group
 - For younger patients: Use WhatsApp with SMS back-up
 - For older patients: Consider phone call reminders in addition to digital reminders
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Insight 6: Evening Appointments Have Longest Wait Times

Finding: Evening appointments average 29.6 minutes waiting time vs. 23.9 minutes in the afternoon and 24.1 minutes in the morning.

Implication: Patient dissatisfaction may be higher for evening appointments due to longer waits, potentially influencing future attendance.

Business Action:

- Review evening appointment scheduling capacity
 - Consider adding additional evening staff
 - Implement a "Wait Time Feedback" system
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Insight 7: Specialist Consultations Show Higher No-Show Rates

Finding: Specialist consultations have the highest no-show rates among all appointment types.

Implication: Specialist appointments often involve longer wait times to schedule and may be perceived as lower priority by patients who are anxious or uncertain about the consultation.

Business Action:

- Implement targeted reminders for specialist consultations
- Provide pre-appointment education to reduce patient anxiety
- Offer a "Specialist Consultation" confirmation call 48 hours prior

6. Business Recommendations

Recommendation 1: Implement a Multi-Channel Reminder System

Action	Details
Priority	High
Timeline	Immediate (0-3 months)
Expected Impact	10-15% reduction in no-shows

Implementation Plan:

1. **Primary Reminder (7 Days Prior):** WhatsApp message with appointment confirmation and address/location details.
2. **Secondary Reminder (3 Days Prior):** WhatsApp/SMS reminder.
3. **Tertiary Reminder (1 Day Prior):** WhatsApp "Final Reminder" with preparation instructions.
4. **Escalation:** If no confirmation from patient, follow with SMS or phone call.

Recommendation 2: Optimise Booking Lead Time Strategy

Action	Details
Priority	High
Timeline	Short-term (0-6 months)
Expected Impact	8-12% improvement in attendance for long lead-time bookings
Implementation Plan:	
Bookings Over 14 days: Automated weekly reminder touchpoint. Bookings Over 30 days: Targeted outreach to confirm attendance interest and offer rescheduling. Waitlist Management: Create a waitlist system to fill slots from cancelled appointments with "Short Term" bookings. Messaging: Reinforce the value and importance of the appointment in all communications.	

Recommendation 3: Address Distance Barriers

Action	Details
Priority	Medium-High
Timeline	Medium-term (3-9 months)
Expected Impact	5-10% improvement in attendance for far-distance patients
Implementation Plan:	

- 1. **Flag System:** Identify Very Far Distance and Far Distance patients in the system.
- 2. **Proactive Outreach:** Contact these patients 48 hours before their appointment to confirm attendance.
- 3. **Telemedicine:** Explore options for teleconsultations for appropriate appointment types.
- 4. **Transport Partnerships:** Consider partnerships with ride-sharing services or community transport.

Recommendation 4: Age-Tailored Communication Strategies

Action	Details
Priority	Medium
Timeline	Medium-term (3-6 months)
Expected Impact	5-8% improvement in attendance for targeted age groups

Implementation Plan:

- 1. **18-34 Age Group:** WhatsApp/Email primary communication, easy rescheduling through digital channels, evening/weekend appointment availability.
- 2. **35-54 Age Group:** Mixed approach (SMS, Email, WhatsApp), flexible scheduling.
- 3. **55+ Age Group:** Phone call reminders in addition to digital, transportation assistance offers, early appointment availability.

Recommendation 5: Improve Appointment Type Management

Action	Details
Priority	Medium-High
Timeline	Short-term (0-3 months)
Expected Impact	5-10% reduction in specialist consultation no-shows

Implementation Plan:

1. **Specialist Consultation:** Dedicated reminder protocol (7, 3, 1 days prior), pre-appointment education, confirmation call 48 hours prior.
2. **Follow-up Appointments:** Book follow-ups at the time of service, provide written appointment details, confirmation via WhatsApp immediately after booking.

Recommendation 6: Optimise Appointment Timing

Action	Details
Priority	Medium
Timeline	Short-term (0-3 months)
Expected Impact	3-5% improvement in patient satisfaction

Implementation Plan:

1. **Evening Appointments:** Review capacity and staffing, implement "Wait Time" monitoring, adjust scheduling to reduce peak times.

- 2. **Morning Appointments:** Prioritize high-risk patients earlier in the day, reduce wait times with better scheduling.
- 3. **Afternoon Appointments:** Maintain current performance, optimise for maximum attendance.

Recommendation 7: Develop a No-Show Prevention Dashboard

Action	Details
Priority	High
Timeline	Short-term (0-3 months)
Expected Impact	Ongoing improvement through data monitoring

Implementation Plan:

- 1. **Key Metrics to Track:** Daily no-show rate by appointment type, no-show rate by reminder status, no-show rate by distance category, no-show rate by booking lead time, no-show rate by age group.
- 2. **Real-time Monitoring:** Identify high-risk appointments, proactive outreach to high-risk patients, daily no-show alert system.
- 3. **Monthly Review:** Evaluate the effectiveness of interventions, adjust strategies based on data, report to management on progress.

7. Dashboard Visualization Summary

The Power BI dashboard developed for this analysis includes the following visualizations:

7.1 Dashboard Tabs

Tab 1: Appointment & Patient Insights

- Booking Lead Days Category and No-Show Rate

- Distance to Clinic Group and No-Show Rate
- Appointment Day and Cancellation/Attendance Breakdown
- Appointment Type and Distance Category Analysis
- Reminder Channel and Attendance Performance
- Attendance Rate by Reminder Status
- No-Show Rate by Reminder Status
- No-Show Rate by Booking Lead Time
- No-Show Rate by Appointment Type

Tab 2: Patient Experience & Process Improvement

- Average Waiting Time by Appointment Time
- Attendance Rate by Booking Lead Days
- Attendance Rate by Distance
- Average Waiting Time by Appointment Type
- Demographic Breakdowns by Age Group

8. Limitations

Data Limitations

Limitation	Description
Missing Values	The reminder_channel column contains 1,366 missing values (27.3% of records), which were left as missing but may represent either no reminder sent or data entry errors.

Limitation	Description
Data Period	The dataset represents a specific time period and may not capture seasonal or long-term trends in appointment behavior.
No Causal Inference	This analysis identifies correlations and patterns but cannot establish causation. Other unmeasured factors may influence attendance.
Self-Reported Data	Appointment outcomes (attended, no-show, cancelled) are assumed accurate but may contain recording errors.

Analytical Limitations

Limitation	Description
Segmentation Granularity	Age group, distance, and lead time categories were pre-defined, and alternative groupings might reveal different patterns.
External Factors	Socioeconomic status, weather conditions, and other external factors are not captured in the dataset.
Generalizability	Findings from this dataset may not be directly applicable to other clinic settings or patient populations.

9. Week 5 Project Summary

What I Planned to Accomplish:

- Clean and prepare the appointment data
- Calculate key KPIs
- Create a Power BI dashboard
- Provide business recommendations

What I Actually Completed:

- Cleaned dataset (handled missing values, standardized data types)
- Calculated all 7 KPIs
- Built 2-page Power BI dashboard
- Identified 7 key insights
- Made 7 practical recommendations

Key Findings or Development Outcomes:

1. No-show rate is 48.5% - a critical problem
2. Current reminders improve attendance by only 1.3% - not effective
3. Booking lead time is a strong predictor of attendance
4. Distance is a major barrier to attendance
5. Different age groups show different patterns

Major Challenges Encountered:

Handling the large number of missing values in reminder_channel (1,366 records)

Important Decisions Made and Why:

- Left reminder_channel missing values as is (represents "Absent of information")
- Replaced missing waiting_time and distance values with "Not Specified"

- Focused on actionable insights rather than just reporting numbers

Any Changes to My Week 4 Approach:

- Added more focus on reminder effectiveness after finding it was not working well
- Included additional KPIs to better understand the problem

Remaining Work:

- Deeper analysis of specialist consultation no-shows
- Cost-benefit analysis of recommendations
- Refine dashboard with additional filters

Proposed Focus for Week 6:

- Predictive modelling to identify high-risk patients at time of booking
- Deeper analysis of why reminders are not working
- Refine recommendations with implementation costs

10. Conclusion

The HealthConnect appointment analysis reveals that **48.5% of scheduled appointments result in no-shows**, representing a significant operational challenge. The evidence shows that:

Finding	Impact
Reminders improve attendance by only 1.3%	Current reminder system needs significant improvement

Finding	Impact
Booking lead time is a critical predictor	Short Term bookings (0-7 days) have 66.6% attendance vs. 34.1% for 1-2 Months bookings
Distance is a major barrier	Very Far Distance patients have 68.1% no-show rates
Younger patients (18-24) show the highest no-show rates	Need tailored communication strategies
Specialist consultations have the highest no-show rates	Need extra attention and targeted reminders
The recommendations provided offer a comprehensive, data-driven approach to reducing no-shows, improving patient engagement, and enhancing the overall HealthConnect patient experience. Through the implementation of multi-channel reminders, proactive outreach to high-risk patients, and targeted strategies by age, distance, and appointment type, HealthConnect can significantly reduce the 48.5% no-show rate and improve clinic efficiency.	

End of Report

Prepared for AnalystLab Africa - HealthConnect Week 5 Assignment

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